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St. Joseph's college of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore–607001
B. Sc DEGREE EXAMINATION APRIL 2021
ABC202-ADVANCED BIOCHEMISTRY

Time : 3 hrs **Max Marks :75**

SECTION–A (25X1=25)
Answer ALL the questions

I choose the best answer: (10X1=10)

1. The bioenergetics of anaerobic glycolysis is

   a) 2 ATP b) 8ATP c) 12 TAP d) 38 ATP

2. The site of glycolysis is

   a) Mitochondria b) Nucleus c)Cytosol d) Golgi apparatus

3. Four digit numbering of enzyme second digit denotes

   a) Main class b) sub class c) sub sub class d) number

4. Preformed active site

   a) Induced fit b) lock and key model c) a and b d) flexible active site

5. In hemolytic jaundice ---------- is increased.

   a) Unconjugated bilirubin b) conjugated bilirubin c) total bilirubin d) virus

6. Gout is a metabolic disease associated with overproduction of

   a)ammonica b) uric acid. c) amino acid d) xanthine

7. All the following statements about albinism are correct except

   a) Tyrosine hydroxylase (tyrosinase) is absent or deficient in melanocytes

   b) Skin is hypopigmented

   c) It results in mental retardation

   d) Eyes are hypopigmented

8. Alkaptonuria occurs due to deficiency of the enzyme
    a) Maleylacetoacetate isomerise
    b) Homogentisate oxidase
    c) p-Hydroxyphenylpyruvate hydroxylase
    d) Fumarylacetoacetate hydrolase
9. Isoenzymes are
    a) Chemically, immunologically and electrophoretically different forms of an enzyme
    b) Different forms of an enzyme similar in all properties
    c) Catalysing different reactions
    d) Having the same quaternary structures like the enzymes
10. The isoenzymes LDH5 is elevated in
    a) Myocardial infarction b) Peptic ulcer
    c) Liver disease d) Infectious diseases

II Say true or false (5X1=5)

11. L-glutamate dehydrogenase can employ NAD+ or NADP+ as Cofactor.
12. The induced fit model of enzyme action was proposed by Koshland.
13. The term hyperglycemia refers to an increase in the blood urea above the normal level.
14. Alkaptonuria occurs due to deficiency of the enzyme Maleylacetoacetate isomerise.
15. Electrolyte are a compound with ionic bonds, forming ions and capable of transmitting electric current in solution.

III Answer all the questions (10X1=10)

16. Name the amino acid which transports ammonia from tissue to liver or kidney.
17. What is the product of condensation of acetyl-CoA with oxaloacetate?

18. Define active site.
19. Give example for competitive inhibition.
20. Give the types of jaundice.
21. Write the classification of diabetes mellitus.
22. Give the defective enzyme in albinism.
23. Give the defective enzyme in people suffering from maple syrup urine disease.
24. Write the random blood glucose level.
25. What is the Daily requirement of calcium for normal adult human?

SECTION -B (4X5=20)

Answer any FOUR of the following

26. Write the reactions of anaerobic glycolysis with its importance.
27. Write short note on classification of enzymes.
28. Give note on the causes and symptoms of type II diabetes mellitus.
29. Explain the causes and symptoms of albinism.
30. Explain the importance of normal blood glucose levels.

SECTION –C (3X10=30)

Answer any THREE of the following

31. Give detailed account on TCA cycle with its bioenergetics.
32. Elaborate on enzyme inhibition
33. Write essay on types, causes and symptoms of jaundice.
34. Explain the inborn error associated with Maple's syrup and Hartnup's disease.
35. Define electrolyte and add note on importance of any three.

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